

Requirements Engineering

Measuring Value

Accelerating Change

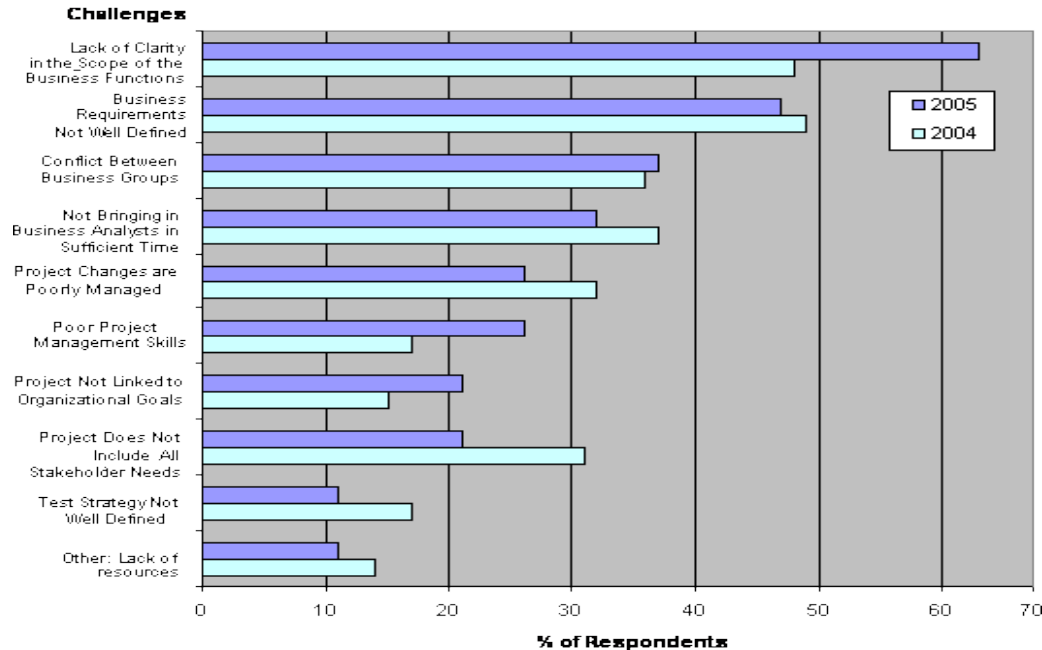
Anticipating Impact

August, 2022

Not Much Has Changed

Top Challenges with Requirements (20 years ago!)

What are the top 3 challenges your organization faces in managing your business requirements?



Source: [Business Improvement Architects](#)

Five common errors in requirements analysis

1. Customers don't (really) know what they want
2. Requirements change during the course of the project
3. Customers have unreasonable timelines
4. Communication gaps exist between customers, engineers and project managers
5. The development team doesn't understand the politics of the customer's organization

Source: <http://www.techrepublic.com/article/five-common-errors-in-requirements-analysis-and-how-to-avoid-them>

Does this sound familiar ?



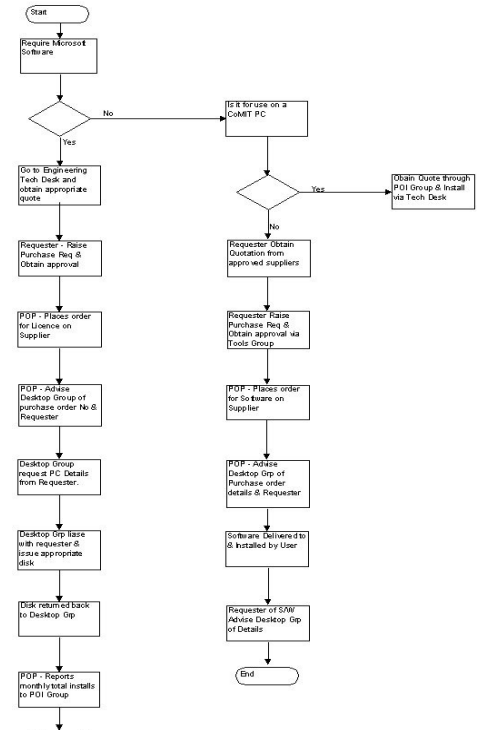
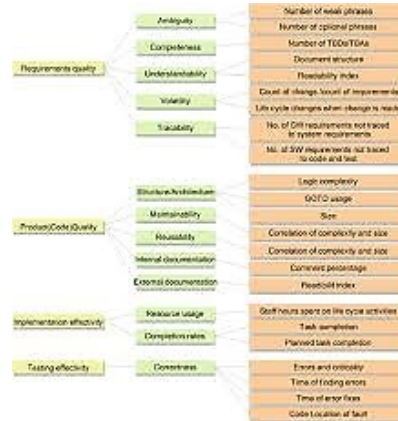
What Makes a Good Requirement?

- complete
- consistent
- correct
- modifiable
- ranked
- traceable
- **unambiguous**
- understandable
- testable

NASA, Automated Requirements Measurement Tool,
1999, <http://satc.gsfc.nasa.gov/tools/arm> (archive.org)

Ambiguity Category	Number of occurrences
Contradiction	7
Ambiguity of reference	10
Dangling Else (including improper use of conditionals)	9
I.E. versus E.G.	1
Omissions (Missing Clauses)	5
Omissions (Missing Definitions)	3
Completely Ambiguous	1
Ambiguous Logical Operators	2
Ambiguous Precedence Relationships	5
Incomplete Definition Completed Elsewhere	3

Total ambiguities found: 46



Unambiguous, Incomplete Requirements

Introduce avoidable uncertainty at the very beginning

- Poor quality requirements introduce uncertainty at the beginning, which is then perpetuated throughout the SDLC
- A system can be built around misunderstanding, so that:
 - At least **56% of defects stem from ambiguity in requirements**
 - **64% of total defect cost**¹originate in the requirements analysis and design phase
 - **40-50% of project costs** are expended in rework²
 - On average, **only 69% of desired functionality is actually delivered**³

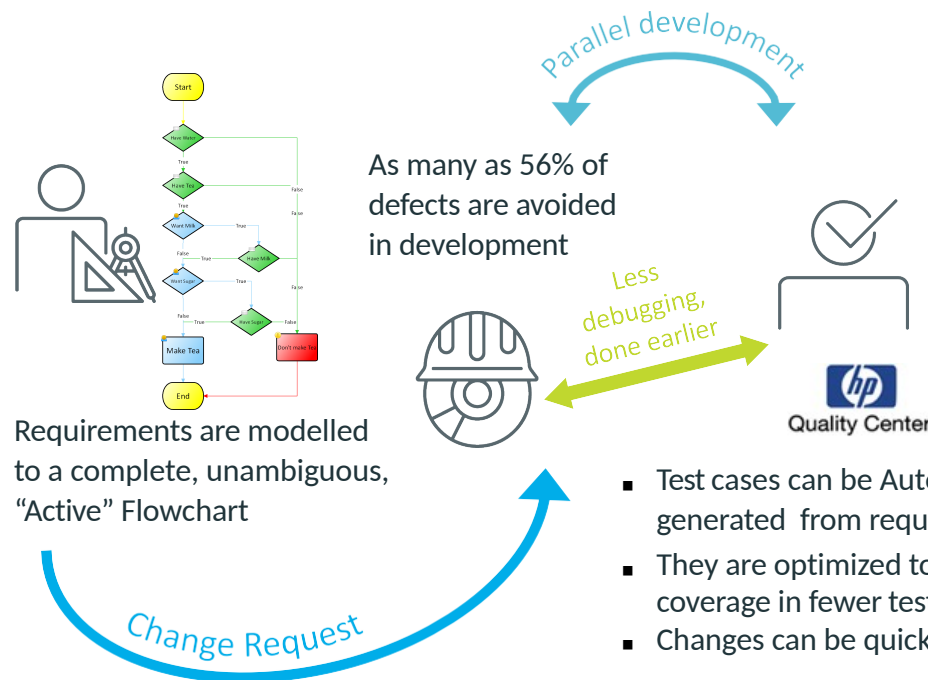
1. Hyderabad Business School, 2012

2. Critical-Logic, 2014

3. Standish Group's Chaos Manifesto 2014

Reduce Uncertainty – Clear Requirements and Fewer “Hops”

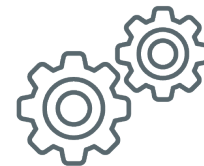
Requirements



- Requirements and test cases are linked to data with 100% coverage, including negative and future scenarios



- Automated tests are generated from test cases
- Are updated if requirements change
- Exhaustively test application
- Pallet-based UI Automation



- Test cases can be Auto-generated and re-generated from requirements
- They are optimized to provide 100% coverage in fewer tests
- Changes can be quickly and easily made

Current State

Today requirements are :

- drafted as documents
- translated into artifacts
- decomposed into features
- captured in doc objects, spreadsheets, and models

Some challenges include :

- subject to interpretation
- not easily reused or repurposed
- difficult to achieve consistency
- quickly becomes irrelevant (shelf-ware)

Future State

Tomorrow requirements could be :

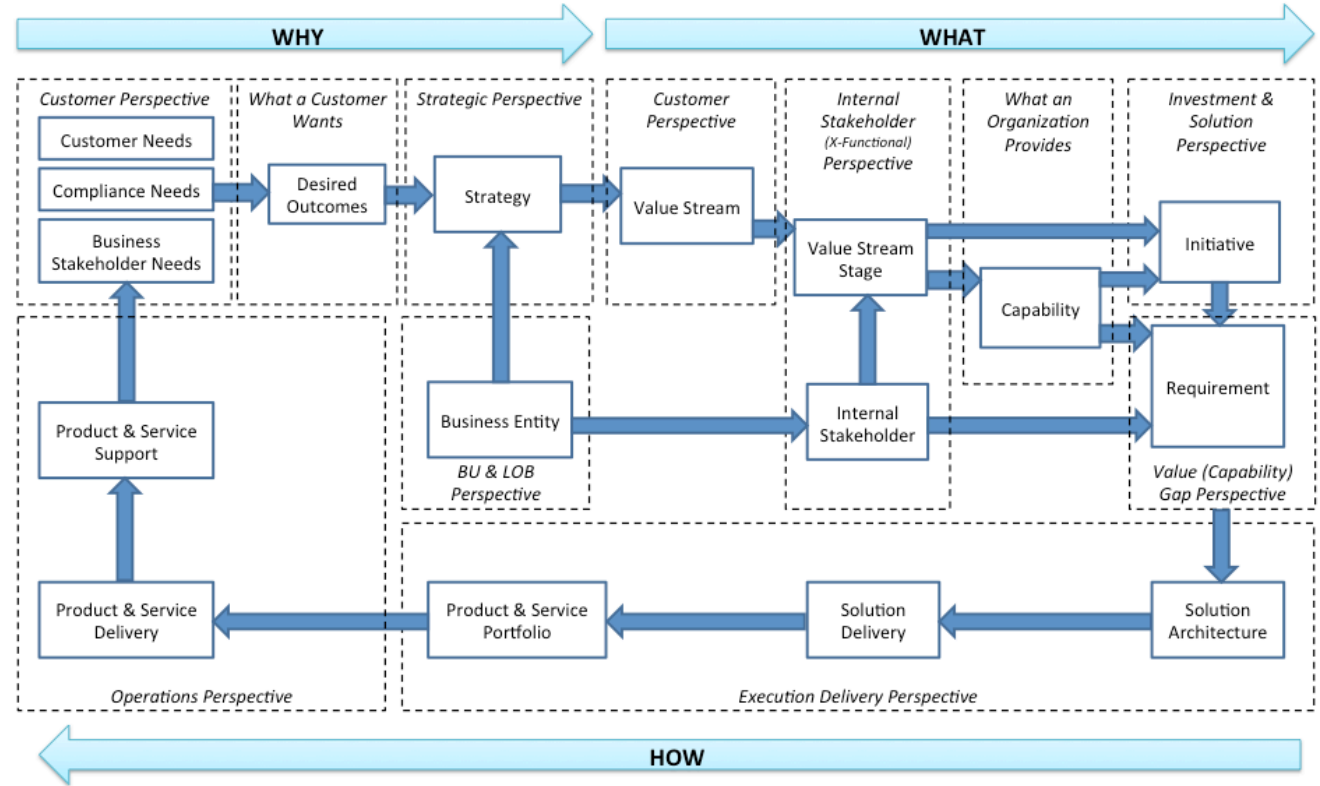
- drafted as capabilities and features which are aligned with services (business & infrastructure)
- captured as business rules, metadata, metrics, and function points within rules libraries, metadata repository, process models, and analytics models

Some of the benefits may include :

- reuse of requirements (all levels of granularity)
- high degree of consistency, relevancy, and usability
- Requirements become part of the solution (subject to version management & part of configuration)

Aligning the WHY, WHAT, & HOW across all POVs

- Requirements are framed and aligned using business architecture to ensure they address the perspectives (POVs).
- When requirements are properly framed, they align strategy and execution to ensure the right value is delivered through the products and services
- Business architecture provides a consistent framework for aligning what customers want against what the organization provides as well as driving continuous improvement in value delivery (by closing the right value & capability gaps)



Data Engineering Systems Engineering Knowledge Engineering **Requirements Engineering**

Definition: Requirement Engineering

ISO/IEC/IEEE 29148:2011

Interdisciplinary function that mediates between the domains of the acquirer and supplier to establish and maintain the requirements to be met by the system, software or service of interest. Requirements engineering is concerned with discovering, eliciting, developing, analysing, determining verification methods, validating, communicating, documenting, and **managing requirements**.



Wikipedia

Refers to the process of defining, documenting and **maintaining requirements** to the sub-fields of systems engineering and software engineering concerned with this process.

What is Requirements Engineering?

Requirements engineering is the use of heuristics for discovering, documenting, and **maintaining** a set of requirements for a system or service

At the highest level, requirements engineering seeks to:

- Define the necessary and sufficient system scope and features to enable system development at an acceptable level of risk
- Enable requirements-driven architecture, design, construction, and verification activities
- Provide information about system development progress and status
- Provide information to organizational learning and continuous improvement efforts

Erik Simmons, nuCognitive, 11/09/16 adapted from “a discussion of the method”, Billy Vaughn Koen, Oxford University Press 2003

Building the Requirements Lineage

Initiative (Problem and/or Project)

- Scenarios (Scope)
 - Requirements
 - Persona
 - Jobs and Job Steps
 - Desired Outcomes (Functional & Emotional)
 - FFs
 - NFFs

- Requirements Cardinality is typically 1:M as you progress from top down
- Initiatives frames the problem domain with specific scenarios
- Scenarios frame the Customer Personas and the Jobs they are trying to get done
- Outcomes provide a granular measure of value (Functional & Emotional Needs)
- Requirements ensure the Design & Build for the Solution delivers the right Outcomes
=> Desired Outcomes for Each Job in scope)
- The Right Solution should deliver Customer Journeys as an orchestrated set of activities to help a Customer complete one or more Jobs (Delivered Outcomes)
- FFs map to value (as measured by Desired Outcomes)
- NFFs used to guide Reuse, Resiliency, Scale and key “ilities”

Feature Taxonomy:

FFs: Functional Features

Typically associated with Business & Functional Services

NFFs: Non-Functional Features

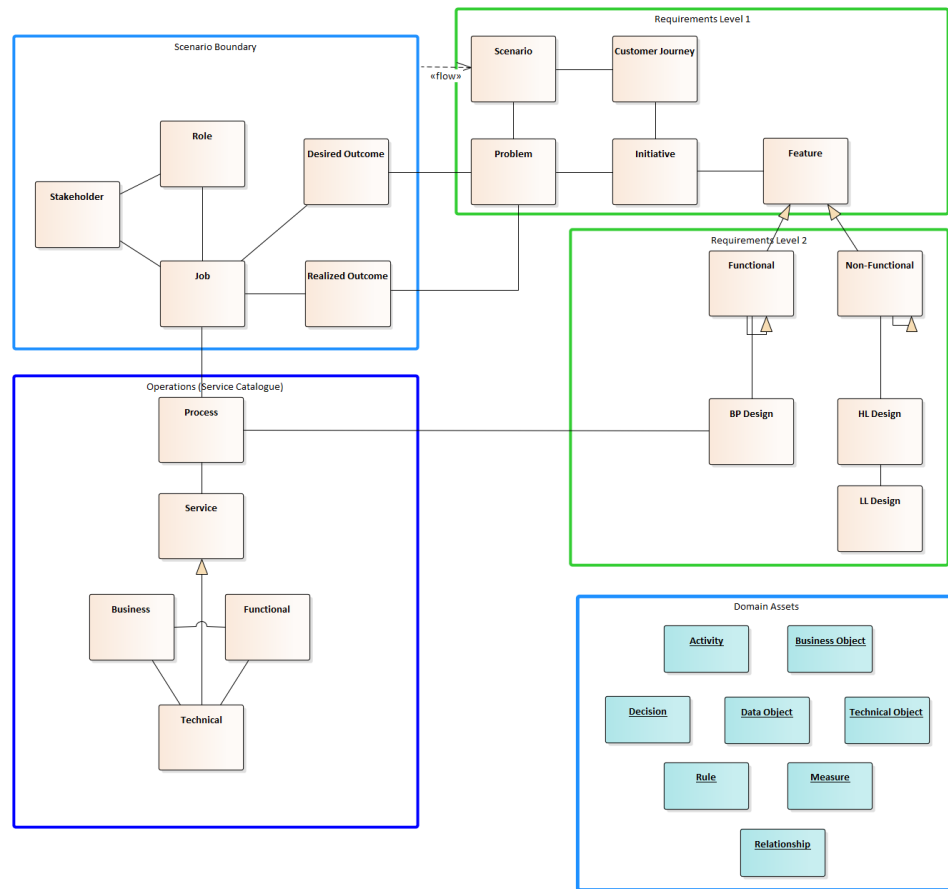
Typically associated with Technology Services

(Architecture Features emphasize the “ilities” & the “itys”)

- Security
- Continuity
- Maintainability
- Usability
- Manageability
- Reliability
- Resilience

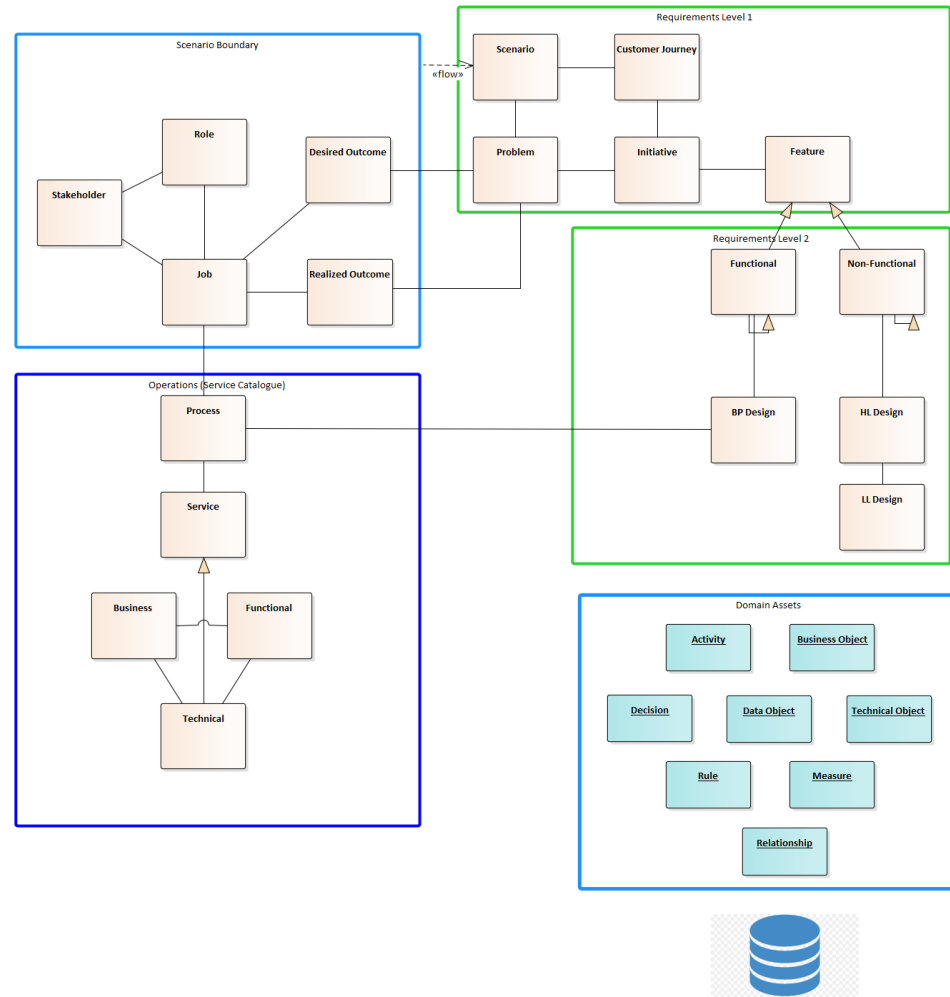
Requirements Modeling

- How do we link all this information together?
- There's a lot for just ONE project
- What about all the other projects?
- How are all of these projects related?
- Where can we drive reuse of requirements?
- Is it enough to model the Requirements?

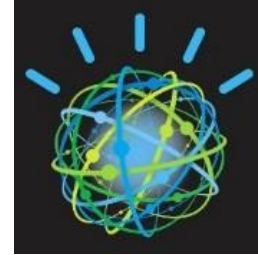


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Data Engineering
Systems Engineering
Knowledge Engineering
Requirements Engineering



**Applying AI to Enable
Cognitive Computing**

What IF...we extend our models beyond Requirements?

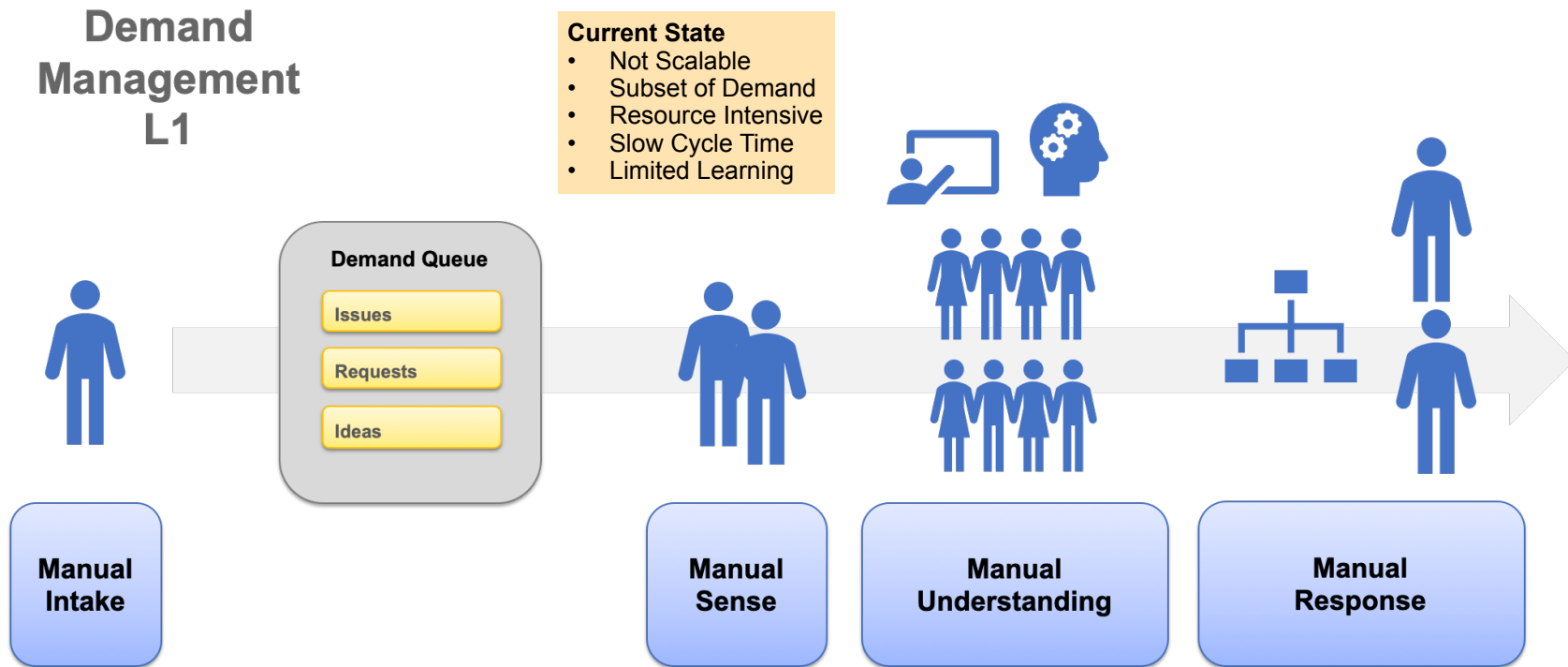
A few questions to consider:

- What if we also modeled the domains and the problem space?
- What if we designed the solutions as reusable services that we can combine to solve one or more problems?

Now, consider the possibilities:

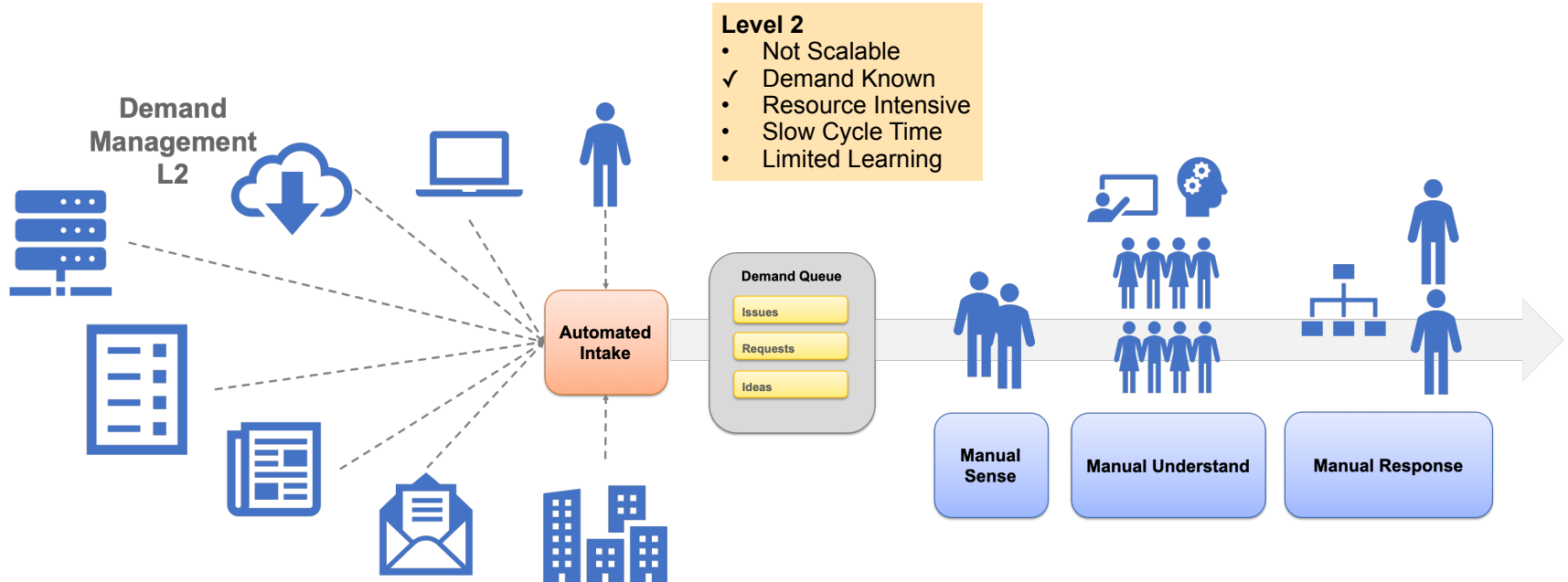
- When we sense a problem that we've seen before (perhaps a specific or similar scenario), perhaps we can apply the same set of services to address the problem (a.k.a. [deliver the right outcomes](#))
- What if we applied AI to help us identify and catalog these patterns?
- Could we apply AI to help us select and orchestrate the right services?

Sensing Change by Modeling the Problem Space



Automating Our Ability to Sense Change

(Intake of Issues & Requests and Mapping them to Requirements "patterns")

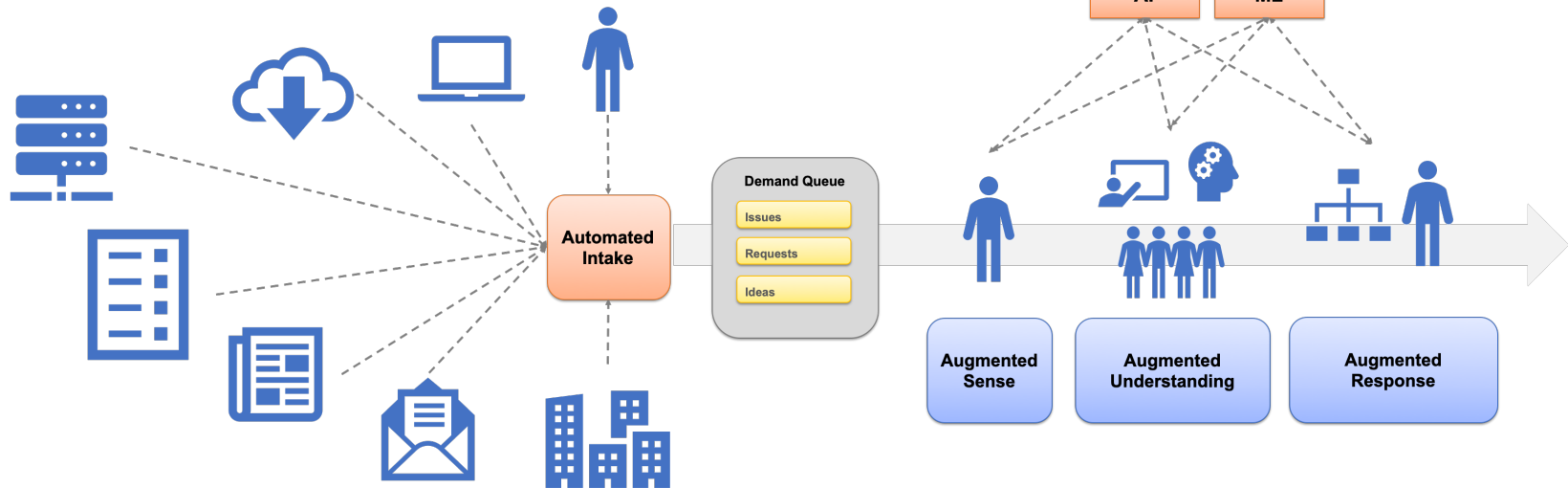


Automating Sense & Response

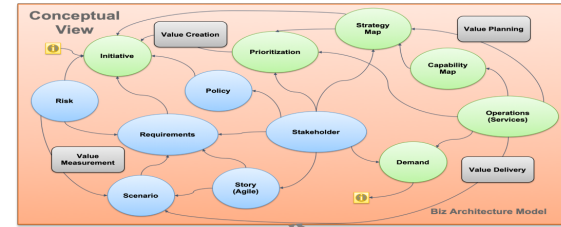
Demand
Management
L3

Level 3

- ✓ Scalable
- ✓ Demand Known
- Resource Intensive
- ✓ Improved Cycle Time
- ✓ Improved Learning



Business Architecture



Enabling Cognitive Computing

Risk Management (Operationalize using Cognitive Computing)

Risk Management

- Incident Management
- Vendor Management
- Compliance Management
- Business Continuity
- Audit Management
- Financial Management
- Policy Management
- Security Management

Operationalizing Risk Management

- Policy Automation & Monitoring
- Threat Monitoring & Remediation
- Vendor SLM monitoring
- Contract Management
- Issue & Incident Analysis
- Continuous Learning

- Sense & Identify
- Contextualize & Understand
- Assess Impact & Evaluate Options (recursive)
- Respond & Monitor (continuous learning)

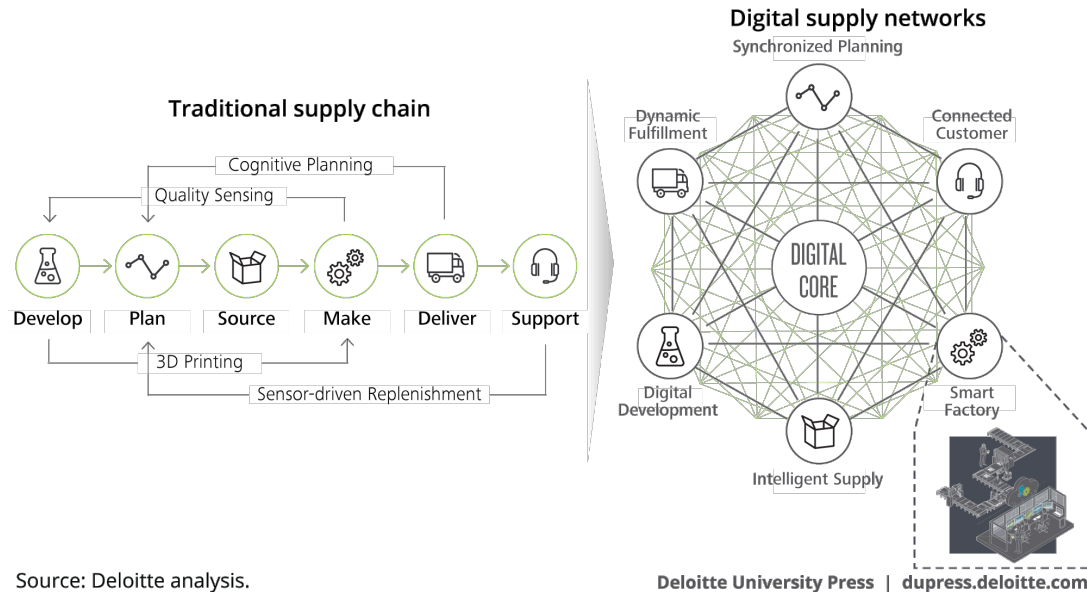


- Identify
- Assess
- Mitigate
- Monitor

Traditional Supply Chain vs. Digital Supply Networks

- Note the multi-faceted network of relationships to support collaborative computing.
- These relationships will be realized and mined from the causal data captured in our EKG.

Figure 1. Shift from traditional supply chain to digital supply network



Supply Chain Use Cases (Cognitive Computing Opportunities)

Rethinking Demand Management

- Today companies try to get very precise on imprecise data. Rows and columns define forecasting. Companies lose visibility on the patterns and demand flows.
- Cognitive learning solutions provide systems of insights that can combine profile pattern recognition along with learning on unstructured data.

Examples include:

- The number of google searches on an illness or symptoms, which is a predictor of the spread of an illness and subsequent prescription sales.
- Social sentiment on twitter and Facebook combined with point-of-sale data drives insights to understand regional sales in days. Today it takes weeks.

Supply Chain Use Cases (Cognitive Computing Opportunities)

Rethinking Planning Master Data

- Only a handful of companies that have a planning master data layer feeding their supply chain planning engines. Most companies install supply chain planning solutions, stabilize the implementation, and then forget about them.
- The data parameters of lead times, cycle times, and rates quickly become outdated. In most cases this data is a variable not a constant.

Examples include:

- Moving rail through Chicago in the winter is not the same as the summer.
- Unloading a container in Long Beach varies by the season; Yet the planning systems have a fixed value.

Supply Chain Use Cases (Cognitive Computing Opportunities)

Rethinking Decision Support

❖ Building Systems of Insights

Cognitive computing, layered on top of existing decision support tools:

- revenue management
- trade promotion management
- demand planning
- production planning
- transportation planning
- supply network planning—drives deeper insights

Supply Chain Use Cases (Cognitive Computing Opportunities)

Establish Sensory Capabilities

- Today's analytics drive answers on the questions we know, but we cannot track the important data that answers the questions which we do not know. Examples are many. Why are consumers raising concerns on quality? What do patterns tell us?

Monitoring & Measuring Quality at Scale & Complexity

- Many production environments—coolers, dryers and distillation columns—are complex with many variables. Once the variables exceed 5 to 7 it is hard to model outcomes without collinearity. Cognitive learning enables new forms of insights to better control quality.

Driving Customer-Centricity in Supply Chains, Products, and Service Delivery

- Cognitive learning is ideal to map customer policies to fulfillment. The dynamic nature of inventory to order matching is difficult. This is an area of great opportunity.

Supply Chain Use Cases (Cognitive Computing Opportunities)

From InteGREAT to InterOPERATE

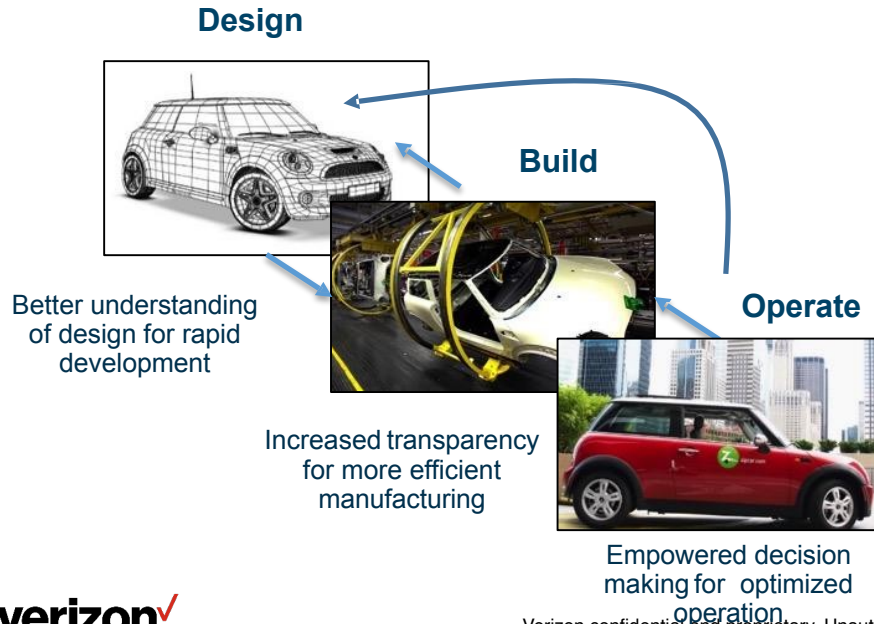
- The connection of data between systems of systems is pre-defined with a fixed set of scenarios.
- Consider replacing the integration with an adaptable messaging layer that uses cognitive learning to guide behaviors between systems and things (IoT).
- Cognitive learning improves data mapping and gives insights to data relationships and inferences which are not obvious.

Putting it all together

Cognitive Digital Twin: Blending Digital and Physical Worlds

Cognitive digital twin:

A virtual representation of a physical object or system across its lifecycle (design, build, operate) using real time data from IoT sensors and other sources to enable learning, reasoning and automatically adjusting for improved decision making.



A comprehensive set of capabilities and information models

- Enable shorter design cycle and speed up innovation
- Support change management and solution impact analysis
- Provide tight linkage between design, build, and operate
- Optimize asset operation & performance
- Leverage operational insights back to design